

Question 8

Q: Given a noisy image acquired from a sensor, explain how the acquisition process contributes to noise and how it can be minimized.

Theoretical Concept: Sensor Noise

During image acquisition, noise is introduced by thermal fluctuations (dark current noise) and electronic circuitry (read noise). This is modeled as additive Gaussian noise or Salt-and-Pepper noise.

Minimization: Can be minimized via hardware (cooling the sensor) or software using Spatial Filters like the **Median Filter** (excellent for salt-and-pepper) or **Gaussian Filter**.

OpenCV Python Solution

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```
import cv2 # Import OpenCV for image processing algorithms
import numpy as np # Import Numpy for mathematical mat
import matplotlib.pyplot as plt # Import Matplotlib to visualize th

# Step 1: Create a clean base image
img = np.full((100, 100), 128, dtype=np.uint8) # Create a solid gray image array filled with a

# Step 2: Introduce Synthetic Salt-and-Pepper sensor noise
noise = np.random.randint(0, 100, (100, 100)) # Generate random numbers to simulate sensor nois
img[noise > 95] = 255 # Salt (White)
img[noise < 5] = 0 # Pepper (Black)

# Step 3: Clean the noise using a Median Filter
# The '5' indicates a 5x5 pixel neighborhood used to calculate the mathematical median
clean_img = cv2.medianBlur(img, 5) # Apply median filter to remove salt-and-pepper noise

# Step 4: Display the cleaned image
plt.figure(figsize=(8,4)) # Create a new figure window for display
plt.subplot(121), plt.imshow(img, cmap='gray'), plt.title('Sensor Noise') # Place this image in
```

```
plt.subplot(122), plt.imshow(clean_img, cmap='gray'), plt.title('Median Filter Applied') # Plac
plt.show() # Show the final plot window to the user
```

Expected Output

The output shows a highly speckled, noisy image next to a perfectly restored, flat gray image. The median filter calculates out extreme outliers without blurring actual structure.

How to Execute & Submit

1. Google Colab Execution:

- Open [Google Colab](#) and create a New Notebook.
- Click the  **Copy** button on the code block above.
- Paste it into a Colab cell and press **Shift + Enter** to run.

2. Uploading to GitHub:

- In Colab, go to **File > Save a copy in GitHub**.
- Authorize your GitHub account.
- Select your Computer Vision Lab repository and click **OK** to commit the code.